

Study Overview

Brief Summary

This is a prospective, non-randomized and multicenter study designed to compare biological features between pN0 triple negative breast cancer (TNBC) with size ≤ 10 mm (pT1a/b) versus pT1c T2 ≤ 30 mm.

All consecutive patients will be recruited by each investigator after completion of surgery. No modification of standard management according to each investigator center will be done. All patients will then be followed each year during 5 years in order to collect the following events: local and loco regional recurrence, metastasis, second cancer, death or not and the cause.

At initial visit, a 10 mL blood sample will be collected (= study intervention) and immediately processed for serum storage; all serum samples will be stored at -80°C and may be used for the purpose of further scientific research.

A representative formalin-fixed paraffin-embedded tumor block of all 200 samples will be addressed at the Institut Claudius Regaud for central collection which will consist of one haematoxylin-eosin stained slide for central histological review, up to 15 unstained slides for DNA extraction (after microdissection), and construction of a tissue micro-array (TMA).

Extracted DNA from 100 samples (50 in each group) will then be transferred to Institut Paoli Calmettes, Marseille; extracted DNA will be subjected to array-CGH analysis in order to detect gene copy number alterations such as gains/amplifications/deletions, and to next generation sequencing (NGS; MiSeq, Illumina) using a panel of ~400 genes for mutation detection.

Contacts and Locations

This section provides contact details for people who can answer questions about joining this study, and information on where this study is taking place.

To learn more, please see the Contacts and Locations section in How to Read a Study Record.

This study has 11 locations

France

Angers, France, 49933

Institut de Cancerologie de L'Ouest - Site Paul Papin

Caen, France, 14076

Centre Francois Baclesse

Clermont-Ferrand, France, 63011

Centre Jean Perrin

Le Puy-en-Velay, France, 43000

CH Emile ROUX

Limoges, France, 87042

CHU de LIMOGES - HOPITAL DUPUYTREN

Marseille, France, 13273

Institut Paoli Calmettes

Montpellier, France, 34298

Institut Regional Du Cancer Montpellier

Quint-Fonsegrives, France, 31130

Clinique La Croix Du Sud

Saint-Herblain, France, 44805

Institut de Cancerologie de L'Ouest - Site Rene Gauducheau

Toulouse, France, 31059

Institut Claudius Regaud

Villejuif, France, 94805

Institut Gustave Roussy

Participation Criteria

Researchers look for people who fit a certain description, called eligibility criteria. Some examples of these criteria are a person's general health condition or prior treatments.

For general information about clinical research, read [Learn About Studies](#).

Eligibility Criteria

Description

Inclusion Criteria:

Woman with age ≥ 18 years.

Patient who completed surgery for his breast cancer and for which definitive histo-pathological analysis of surgical specimen is available.

Invasive breast carcinoma pN0 or pN(i+) with histological tumor size ≤ 10 mm (pT1a/b subgroup) or invasive breast carcinoma with histological tumor size > 10 mm and ≤ 30 mm (pT1c T2 ≤ 30 mm control group).

Patient with HER2-negative breast carcinoma: immuno-histochemistry (IHC) score = 0, 1+ or 2+ and in situ hybridization (FISH, CISH, or SISH) negative (local laboratory testing).

Patient with ER and PR negative invasive carcinoma ($< 1\%$ stained cells by immuno-histochemistry assay) (local laboratory testing).

In case of multifocality, the histological size of the largest tumor must be ≤ 10 mm or ≤ 30 mm according to the inclusion subgroup. All lesions must be ER, PR and HER2-negative.

In case of breast conserving surgery, clear margins are required.

Patient affiliated to a Social Health Insurance in France.

Patient information and written informed consent form signed prior to any study specific procedures.

Exclusion Criteria:

Patients with any previous malignancy of the breast or other site, with the exception of adequately treated basal or squamous cell carcinoma, non-melanomatous skin cancer totally resected.

Non-invasive breast carcinoma (i.e. ductal carcinoma in situ exclusively).

Inoperable breast invasive carcinoma.

Synchronous bilateral breast cancer.

Patients who received neo-adjuvant treatment (radiotherapy or chemotherapy or other before surgery).

Pregnant or breast-feeding women.

Any psychological, familial, geographic or social situation, according to the judgment of investigator, potentially preventing the provision of informed consent or compliance to study procedure.

Patient who has forfeited his/her freedom by administrative or legal award or who is under guardianship.

Study Plan

This section provides details of the study plan, including how the study is designed and what the study is measuring.

[Expand all](#) / [Collapse all](#)

How is the study designed?

Design Details

Primary Purpose : Other

Allocation : Non-Randomized

Interventional Model : Parallel Assignment

Masking : None (Open Label)

Arms and Interventions

Participant Group/Arm Intervention/Treatment

Other: TNBC pT1a/b with size $< \text{or} = 10$ mm

Patients with pN0 triple negative breast tumor with size $< \text{or} = 10$ mm (pT1a/b)

Study intervention = blood samples collected at initial visit

Procedure: Blood samples of 10 mL

Blood samples of 10 mL will be collected at initial visit; serum samples will then be stored at the end of the study (-80°C) and may be used for the purpose of further scientific research.

Other: TNBC pT1c T2 with size $< \text{or} = 30$ mm

Patients with pN0 triple negative breast tumor with size ≤ 30 mm (pT1c T2)
Study intervention = blood samples collected at initial visit
Procedure: Blood samples of 10 mL
Blood samples of 10 mL will be collected at initial visit; serum samples will then be stored at the end of the study (-80°C) and may be used for the purpose of further scientific research.

What is the study measuring?

Primary Outcome Measures

Outcome Measure	Measure Description	Time Frame
Biomarker signature for pT1a/b pN0 triple-negative breast cancers versus pN0 triple-negative breast cancers with size ≥ 11 mm and ≤ 30 mm.	biomarker signature will be defined from the protein expression of one or several biomarker(s).	3 years.

Secondary Outcome Measures

Outcome Measure	Measure Description	Time Frame
Disease free survival.	Disease Free Survival is defined as the time from inclusion until any recurrence (local or regional), contralateral breast cancer, second primary breast cancer, distant metastasis, or death from any cause.	7 years.
Metastasis Free Survival.	Metastasis Free Survival is defined as the time from inclusion until distant metastasis or last follow-up news (censored data). All other events are ignored for this endpoint.	7 years.
Overall survival.	Overall survival is defined as the time from inclusion until death from any cause or last follow-up news (censored data).	7 years.
Description and comparison of gene copy number (array-CGH) and gene mutation (targeted NGS) between pT1a/b pN0 triple-negative breast cancer versus pN0 triple-negative breast cancer with size ≥ 11 mm and ≤ 30 mm.		7 years.

Collaborators and Investigators

This is where you will find people and organizations involved with this study.

Sponsor

Institut Claudius Regaud

Investigators

Study Chair: Florence DALENC, Md., Institut Claudius Regaud

Study Record Dates

These dates track the progress of study record and summary results submissions to ClinicalTrials.gov. Study records and reported results are reviewed by the National Library of Medicine (NLM) to make sure they meet specific quality control standards before being posted on the public website.

Study Registration Dates

First Submitted

2014-10-13

First Submitted that Met QC Criteria

2014-10-16

First Posted (Estimated)

2014-10-17

Study Record Updates

Last Update Submitted that Met QC Criteria

2024-12-02

Last Update Posted

2024-12-03

Last Verified

2024-08